Computer Networks 1B Revision Guide

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This guide explains all 45 learning outcomes across the six learning units in your module outline. The explanations and examples are written for revision: first understand the idea, then practise using it to solve a problem. The scenarios are fictional teaching examples. Use this guide alongside lectures, Arc activities and the prescribed textbook; it does not replace them or predict examination questions.

How to use this guide

Read one outcome at a time. Explain it aloud without looking at the notes. Work through the example, then change a number or a condition and try again. Answer the practice questions before reading the guidance. In an examination, connect every technical point to the scenario: say what the concept means, how it works, and why it matters in that situation.

What the four module outcomes require

MO1 Understanding TCP/IP: explain which layer and protocol performs each task. Example: DNS finds an address, TCP manages a reliable byte stream, IP forwards packets between networks and Ethernet carries frames on a local link.

MO2 Applying configuration practices: select suitable protocols, plan addresses and topology, configure consistently and verify the result. Example: give each subnet a unique range, configure the correct gateway and check DNS separately from basic IP connectivity.

MO3 Solving network problems: identify a cause, choose a suitable remedy and explain the tradeoff. Example: adding bandwidth can help a congested uplink, but it will not correct an invalid gateway or weak Wi-Fi signal.

MO4 Applying security principles: reduce unnecessary exposure, control access and protect management traffic. Example: isolate student and administration devices, allow only required connections, use secure management and maintain updates.

Your revision route

Unit 1 covers the Internet, performance, layers and mobility. Unit 2 covers applications. Unit 3 covers transport. Unit 4 covers forwarding and IP addressing. Unit 5 covers routing and management. Unit 6 covers local delivery and LANs. The outline links these units to textbook Chapters 1 and 7, 2, 3, 4, 5 and 6 respectively; the Unit 2 table labels its reading as Module 2. Consult Arc for the exact reading sections.

1 Computer networks and wireless mobility

LO1 The Internet as a network of networks

The Internet connects independently managed networks through routers and communication links. A host, or end system, runs applications: examples include a laptop, phone, server and smart camera. Switches commonly connect devices inside a LAN; routers forward IP packets between networks. An Internet service provider connects a customer network to other networks. Providers exchange traffic through peering arrangements and Internet exchange points or purchase transit from other providers.

A protocol is an agreed set of message formats, message sequences and actions. HTTP defines web requests and responses; TCP provides a reliable ordered byte stream; IP carries addressed packets; DNS resolves names. An application service such as web browsing uses several protocols together. No single device or organisation owns the whole Internet.

Example: a student opens a learning portal. The laptop connects to a Wi-Fi access point, then the campus router, an ISP and networks serving the portal. DNS helps obtain the server address. The browser exchanges application messages over transport and network services. The return traffic need not take exactly the same route.

Remember: the Internet is infrastructure; the Web is one service using that infrastructure. Email, voice calls and file transfers also use it.

LO2 Access networks

An access network connects an end system to its first router on the path into a wider network. Residential options include fibre, cable, DSL and fixed wireless. Businesses and campuses commonly use Ethernet and Wi-Fi behind a shared Internet connection. Cellular networks provide radio access through a mobile operator. The appropriate choice depends on coverage, capacity, latency, mobility, cost and reliability.

Fibre uses light in optical cable and supports high capacity over substantial distances. DSL uses telephone copper, with performance affected by line length and quality. Cable access shares capacity among subscribers in an area. Ethernet offers stable wired local access. Wi-Fi shares radio airtime and is affected by interference, obstacles and distance. Cellular performance varies with signal conditions and cell load.

Example: a lecturer delivering classes from a fixed office may benefit from fibre and wired Ethernet, with cellular backup. A student travelling between towns needs mobile coverage. A fast campus Wi-Fi connection does not guarantee fast downloads if the campus Internet uplink is congested.

Exam approach: recommend a technology and justify it using the user's needs. A statement such as 'fibre is best' needs evidence about that particular scenario.

LO3 Packet switching and circuit switching

Packet switching divides data into packets that share links with other traffic. Each router receives and forwards packets toward the destination. In store-and-forward operation, a router normally receives a whole packet before transmitting it on the next link. Statistical multiplexing allocates link use according to current demand rather than permanently reserving a portion for each user. This is efficient for bursty traffic but can introduce queues and loss.

Circuit switching reserves resources along a path for a session. Frequency division multiplexing divides capacity into frequency bands; time division multiplexing assigns repeating time slots. A reserved circuit offers predictable allocated capacity, but that capacity can be idle when the user has nothing to send. Circuit switching does not mean every session requires a separate physical cable.

Example: ten learners each occasionally send small messages over one link. Packet switching lets active learners use spare capacity. If all ten begin large downloads, they compete for that link and delay may increase. A circuit arrangement reserves each learner a share whether or not they are currently sending.

Key comparison: packet switching shares resources dynamically; circuit switching reserves them. Internet voice applications typically use packets even though they carry conversations.

LO4 Delay loss and throughput

Processing delay is time spent examining a packet. Queuing delay is time waiting for transmission. Transmission delay is the time needed to place all packet bits onto a link. Propagation delay is the time for a signal to travel along the medium. These are different causes and require different remedies.

Transmission delay = L/R, where L is packet length in bits and R is link rate in bits per second. Propagation delay = d/s, where d is distance and s is propagation speed. At one node, nodal delay is the sum of processing, queuing, transmission and propagation delays. For a multi-hop path, add the relevant components for each hop. Packet loss occurs when packets are discarded, for example because a buffer is full; corruption and other conditions can also cause loss.

Worked example: a 1,500-byte packet contains 12,000 bits. On a 10 Mb/s link, transmission delay = 12,000/10,000,000 = 0.0012 seconds = 1.2 ms. Over 200 km at 200,000,000 m/s, propagation delay = 200,000/200,000,000 = 0.001 seconds = 1 ms. With 0.2 ms processing and 3 ms queuing, total nodal delay is 5.4 ms.

Throughput is the achieved transfer rate. A path with 100, 20 and 50 Mb/s links has an ideal single-flow bottleneck limit of 20 Mb/s; sharing and overhead usually reduce application throughput. Goodput counts useful application data and excludes retransmissions and protocol overhead. A 10 MB file contains 80 Mb, so its ideal transfer time at 20 Mb/s is 4 seconds.

Traffic intensity can be approximated as La/R when arrivals average a packets per second. As utilisation approaches capacity, queues may grow sharply; actual delay depends on traffic patterns. Mistakes to avoid: confusing bytes with bits, transmission with propagation, or bandwidth with achieved throughput.

LO5 Layers and encapsulation

Layering divides networking into responsibilities. In the five-layer teaching model, application handles services such as HTTP and DNS; transport handles process communication such as TCP and UDP; network handles IP addressing and forwarding; link handles local frames; physical carries signals. The four-layer TCP/IP model combines link and physical into network access. The seven-layer OSI model also distinguishes session and presentation functions.

Encapsulation means adding control information as data moves down the stack. Application data becomes a TCP segment or UDP datagram, then an IP packet, then a link frame, then signals. The receiver removes the corresponding information as data moves upward. Each layer uses the service below it and provides a service above it.

Example: a browser sends a request. TCP adds ports and reliability information; IP adds source and destination addresses; Ethernet adds local source and destination MAC addresses and an error-detection trailer. A router removes the incoming link frame, processes the IP packet and creates a new link frame for the next link. Link addresses therefore change across routed hops.

Why it matters: standard interfaces let different hardware and software interoperate. Layering itself does not guarantee delivery; particular protocols, such as TCP, supply reliability mechanisms. Diagnose faults by layer: a disconnected cable differs from failed DNS resolution.

LO6 Wireless networks and links

Wireless links carry signals through radio rather than a cable. Infrastructure Wi-Fi uses access points that connect wireless stations to a distribution network. Ad hoc networks can allow direct station communication without an access point. A personal area network connects nearby personal devices, such as Bluetooth earbuds and a phone. Cellular networks support wider-area coverage through base stations and operator infrastructure.

Radio signals weaken with distance and obstacles. Other transmitters can create interference, while reflections can produce multipath effects. Wireless devices commonly share a medium, so extra active users consume airtime. Link rate and actual useful throughput differ because of contention, retransmission and protocol overhead. Wireless does not automatically mean mobile: a fixed outdoor wireless bridge is wireless but stationary.

Example: 30 students join an access point. Video performance deteriorates even though the wired Internet connection is underused. Possible causes include airtime competition, interference or poor signal. Review access-point placement, channel usage, coverage and capacity rather than immediately buying a faster Internet package.

Security application: authenticate users, encrypt radio traffic where supported and isolate guests from sensitive systems. Coverage and access security are separate design issues.

LO7 Code Division Multiple Access

CDMA allows users to share the same time and frequency resources by assigning spreading codes. A transmitter represents each data bit using a faster sequence of smaller signal elements called chips. The receiver correlates the received mixture with the desired user's code to recover that user's contribution. Suitable codes have low cross-correlation, allowing signals to be separated under appropriate conditions.

Worked teaching example: user A has code [1, 1] and user B has [1, -1]. A sends +1, producing [1, 1]. B sends -1, producing [-1, 1]. Their combined signal is [0, 2]. Correlating with A gives (0 x 1 + 2 x 1)/2 = +1. Correlating with B gives (0 x 1 + 2 x -1)/2 = -1. Each user's bit is recovered in this idealised synchronous example.

The near-far problem occurs when a nearby transmitter's stronger signal overwhelms weaker signals. Power control helps manage received signal levels. Real systems must also deal with noise, imperfect synchronisation and multipath; the simple example assumes ideal codes and timing.

Remember: CDMA separates users by codes, FDM by frequency and TDM by time. Do not treat CDMA as the access method for every wireless technology.

LO8 Architecture of 4G LTE and 5G

A cellular system includes user equipment, a radio access network and a core network. In LTE, user equipment communicates with an eNodeB. The evolved packet core includes functions such as the mobility management entity for control, a serving gateway and a packet data network gateway for user traffic, and subscriber information functions. The network authenticates a subscriber, establishes connectivity and carries IP traffic toward external networks.

In a standalone 5G architecture, a gNodeB provides radio access. Core functions include the access and mobility management function, session management function and user plane function. Their responsibilities separate mobility control, session control and packet handling. Non-standalone 5G deployments can use LTE infrastructure as an anchor, so all 5G deployments do not have identical architecture.

Example: a student joins a video class over a phone. The phone accesses a base station; the operator authenticates the subscription, manages the session and forwards traffic through its core. The application still relies on IP and application protocols above the cellular access system.

5G supports designs for high data rates, low latency and large device populations, with capabilities such as network slicing. These are service and deployment capabilities, not guaranteed performance for every user. Coverage, spectrum, cell load and backhaul affect results. Global connectivity also requires operator interconnection and roaming arrangements.

LO9 Mobility and handover

Mobility management keeps track of a device's location and helps maintain communication as it changes its point of attachment. Handover transfers an active connection between access points or base stations. The system measures radio conditions, selects a suitable target, prepares resources and changes the traffic path. Registration and tracking allow a network to locate a device when incoming communication arrives.

Example: a passenger participates in a call while a car moves between cellular coverage areas. The network must redirect traffic to the next base station. If target resources are unavailable, signals deteriorate too quickly or transfer takes too long, the call may briefly freeze or drop. Packet loss, delay and out-of-order delivery can occur during the transition.

In LTE, mobility control includes the MME and base-station procedures. In standalone 5G, functions such as the AMF support access and mobility management, with session and user-plane functions cooperating to maintain service. Mobility also raises authentication, signalling and location-privacy concerns.

Remember: wireless describes the link; mobility describes changing attachment or location. Moving to a different IP network can affect ongoing connections unless mobility support or application recovery handles the change.

LO10 Mobile IP

Mobile IP aims to let a mobile node remain reachable through a stable home address while attached to another network. In the classical Mobile IPv4 model, a home agent maintains a binding between the mobile node's home address and its care-of address. The care-of address identifies its current attachment. The home agent can intercept packets addressed to the home network and tunnel them toward that location.

Example: a device normally addressed on network A travels to network B. A correspondent keeps sending to the home address. The home agent forwards the traffic through a tunnel to the care-of address. Encapsulation adds an outer header so the packet can cross the intervening network while retaining its original inner addressing.

Advantages include stable reachability and support for continuity without every correspondent immediately learning the new location. Limitations include registration overhead, extra tunnel headers, security requirements and potentially inefficient triangular routing. The home agent may also become a performance or availability concern. Route optimisation and IPv6 mobility mechanisms differ from the simplest Mobile IPv4 model.

Evaluation: Mobile IP is a useful architectural concept, but do not assume ordinary smartphone mobility uses classic Mobile IPv4. Cellular systems have their own mobility and session-management procedures.

Unit 1 practice

1. A 2,000-byte packet crosses a 4 Mb/s link. Calculate transmission delay and explain whether doubling link speed changes propagation delay.

2. Compare packet and circuit switching for a class where learners send messages intermittently.

3. A call breaks while a user moves between cells. Explain two possible causes and the purpose of handover.

Unit 1 answer guidance

1. L = 16,000 bits; L/R = 16,000/4,000,000 = 0.004 s = 4 ms. Doubling the link rate reduces this to 2 ms; propagation delay remains governed by distance and signal speed.

2. Packet switching shares spare capacity among active learners but can queue packets during bursts. Circuit switching reserves a share for each session, which offers predictability but wastes unused reserved capacity.

3. Examples include weak target-cell signal and insufficient target resources. Handover moves the active connection and traffic path to a suitable new base station; poor timing can cause interruption.

2 The application layer

LO1 Application architectures and process communication

An application runs as one or more processes on hosts. A client usually initiates communication; a server listens for requests. In client-server architecture, a service such as a learning portal is hosted by managed servers. Servers can be replicated across many machines, so client-server does not necessarily mean one physical server. In peer-to-peer architecture, peers can both request and supply resources.

Processes use sockets as interfaces to transport services. To contact a process, an application needs an address for its host and a port for the service. For example, the same server can offer web and email services using different ports. An application protocol defines message format and behaviour, such as what counts as a request and what response follows.

Example: a student browser requests a PDF from the portal server. The browser is the client and the portal is the server. In a P2P file-sharing system, the student might obtain different pieces from different peers and upload pieces to others.

Comparison: client-server simplifies central administration and access control; P2P distributes work but introduces challenges involving peer availability, trust and coordination. Explain both the process role and where the resources are hosted.

LO2 Choosing transport services

Applications differ in their needs. A file transfer needs complete, correctly ordered data. A live conversation values timely playback and may tolerate small losses. A service may need high throughput, low latency, confidentiality or a combination. Transport selection must follow those requirements.

TCP provides connection-oriented, reliable, ordered byte-stream delivery, with flow and congestion control. It handles loss through acknowledgements and retransmission. Waiting for missing data can delay delivery of later bytes. UDP provides connectionless datagrams without built-in retransmission, ordering, flow control or congestion control. An application can build additional mechanisms above UDP. Neither TCP nor UDP alone provides encryption.

Example: losing a paragraph in an assignment upload is unacceptable, so reliable delivery matters. In a live voice stream, retransmitting a late audio sample may be less useful than playing newer audio. Many real-time systems therefore use UDP-based protocols with their own media handling. Ordinary DNS queries often use UDP, while DNS also uses TCP when needed.

Remember: UDP is not automatically faster in every application; TCP is not a guarantee of a particular delay or rate. Secure transport can use TLS with TCP or protocols such as QUIC over UDP.

LO3 HTTP DNS and HTTP version features

HTTP follows a request-response model. A request includes a method, resource target, headers and sometimes a body. A response includes a status code, headers and possibly content. GET retrieves a representation; POST submits data for processing. Typical response codes include 200 for success and 404 when the requested resource is not found. HTTPS protects HTTP communication using TLS.

HTTP/1.1 supports persistent connections. HTTP/2 uses binary framing, header compression and multiple logical streams over a connection, improving efficiency for many resources. HTTP/2 still relies on TCP and can experience transport-level blocking when TCP data is lost. HTTP/3 uses QUIC and is a useful extension topic; it is not the same protocol as HTTP/2.

DNS translates names into records through a distributed hierarchy. A client normally asks a recursive resolver, which can use cached information or consult root, top-level-domain and authoritative servers. A records map names to IPv4 addresses; AAAA records to IPv6 addresses; CNAME records identify aliases; MX records identify mail exchangers. DNS has many functions beyond web browsing.

Example: the browser resolves portal.example to an address, connects to the server and requests /notes.pdf. If the name fails but a known reachable IP works, investigate DNS. If the server returns 404, connectivity may work while the requested resource is missing. These are different failures.

LO4 Cookies caching and web performance

A cookie is a small value stored by a browser and returned with matching requests. It can support session identification, preferences and tracking. HTTP requests do not inherently remember previous interactions, so applications use mechanisms such as session cookies to associate requests with a logged-in user. A session cookie usually identifies a session; it need not contain the user's password or all account data.

A cache stores a reusable copy of content. Browser caches avoid repeat downloads on one device; intermediary and CDN caches can serve many users. Freshness information controls reuse. A conditional request can ask whether a stored representation has changed, allowing a 304 response when suitable. Benefits include lower delay, less repeated traffic and reduced origin-server load. Risks include stale content and exposure of private content through inappropriate caching.

Example: a portal logo can be cached because many learners repeatedly request it. Personal marks require careful cache policy because one learner must not receive another learner's result. A login cookie should be protected with suitable settings such as Secure and HttpOnly; tracking also raises privacy concerns.

Evaluation: improve performance through appropriate caching, smaller resources and efficient connections, while preserving correctness and privacy. Cookies identify state; caches reuse content. They solve different problems.

LO5 Streaming DASH and CDNs

HTTP streaming delivers media in chunks that can be played before the entire file has arrived. A player maintains a buffer so brief variations in delivery do not immediately interrupt playback. A large buffer can reduce stalls but also delay the start or increase live latency.

Dynamic Adaptive Streaming over HTTP, or DASH, makes media available in segments at several encoded bitrates. A manifest describes the available choices. The client selects a suitable representation for each segment based on estimated throughput, buffer state and other factors. If available throughput falls, it can choose a lower bitrate to maintain playback. The bitrate is the data rate of the encoded content, not the resolution by itself.

A CDN places copies or delivery servers in multiple locations. Requests can be directed to a suitable server using location, routing, load and service policies. A nearby or well-connected

server can lower delivery delay and reduce long-distance repeated traffic. 'Nearest' is not always geographically nearest.

Example: a class video offers 0.8, 1.5 and 3 Mb/s versions. With fluctuating 2 Mb/s throughput, the player may choose 1.5 Mb/s to leave headroom rather than repeatedly stall at 3 Mb/s. A CDN serves the segments from a suitable regional node, reducing load on the original host.

LO6 TCP socket programming

The TCP server creates a socket, binds it to a local address and port, listens for connections and accepts a client. Accept returns a connected socket used for that client, while the listening socket remains available for more connections. The client creates a socket and connects to the server address and port. The two processes then send and receive bytes and eventually close the connection.

TCP is a byte stream, so one send operation does not necessarily correspond to one receive operation. Applications need message framing: for example, a newline terminator, a fixed length or a length prefix. They must handle partial reads, disconnections, timeouts and invalid input. Large services also need a way to support concurrent clients, such as threads, processes or asynchronous operations.

Teaching example: an echo client sends 'Hello' followed by a newline. The server accumulates bytes until it finds the newline, then sends back that complete message. If the first receive returns 'He' and the second returns 'llo' plus newline, the application must combine them. This is valid TCP behaviour.

Simplified sequence: server socket then bind then listen then accept; client socket then connect; both exchange framed data then close. Test a classroom exercise on localhost or an authorised lab network. Reliability at transport level does not remove the need for correct application logic.

LO7 Architecture and scalability

Scalability is the ability to handle growth without unacceptable performance or management cost. In client-server systems, more clients can increase server processing, storage and outgoing bandwidth demand. Load balancing, replication and caching can spread this work. Central management helps enforce consistent permissions but requires planning for service failure and bottlenecks.

P2P systems distribute resources among participants. Extra peers can add upload capacity as well as demand, which can improve distribution at scale. However, peers may leave unpredictably, use slow connections or provide untrusted data. Discovery, integrity checking and incentives become important. Hybrid systems combine central coordination with distributed delivery.

Worked example: a 100 MB file equals 800 Mb. Sending a separate copy to 100 learners requires 80,000 Mb of server upload. At a 100 Mb/s uplink, the ideal lower bound from server upload capacity is 800 seconds, ignoring overhead and other constraints. Caching reduces repeat origin traffic; P2P can allow recipients to redistribute pieces, but the actual time depends on aggregate upload and participation.

Exam approach: identify where work is concentrated, what grows as users increase, and how the architecture addresses it. Neither architecture guarantees infinite capacity or immunity from failure.

Unit 2 practice

1. Explain the sequence from entering a portal name to receiving a page, using DNS, transport and HTTP.

2. A video buffers on a variable connection. Explain how DASH and a CDN could help.

3. Explain why a TCP receiver must not assume one receive equals one complete message.

Unit 2 answer guidance

1. Resolve the name, establish the appropriate transport and security session, send an HTTP request and receive an HTTP response. Cached DNS or reused connections can shorten the sequence.

2. DASH can choose lower-bitrate segments as throughput falls; a CDN can improve delivery paths and spread server load. Neither fixes every local signal or capacity problem.

3. TCP preserves byte order rather than application message boundaries. Define framing and accumulate partial data until the complete message is available.

3 The transport layer

LO1 Transport layer responsibilities

The network layer delivers packets between hosts; the transport layer delivers data between application processes on those hosts. Transport functions can include segmentation, ports, reassembly, error detection, reliability, flow control and congestion control, depending on the protocol. It is usually implemented at the end systems rather than by every transit router.

Example: one laptop runs a browser and a video application. IP delivers incoming packets to that laptop. Transport processing identifies which process should receive each payload. TCP additionally recovers missing data and delivers bytes in order; UDP delivers individual datagrams without those guarantees.

In OSI and TCP/IP descriptions, transport sits above network and below application-related functions. It builds on IP's best-effort delivery. IP can lose, duplicate or reorder packets, so reliable transport must manage these possibilities rather than assume a perfect underlying network.

Evaluation: TCP's reliability is valuable for documents and transactions but can add delay when recovery is needed. UDP gives applications more direct control over timing but shifts responsibility for required reliability and congestion behaviour to higher-level designs.

LO2 Multiplexing and demultiplexing

Multiplexing at the sender gathers data from different application sockets and creates transport packets with identifying information. Demultiplexing at the receiver uses that information to deliver data to the correct socket. Port numbers identify transport endpoints within a host. They are different from IP addresses, which identify network-layer interfaces or destinations.

TCP connections are distinguished by source IP, source port, destination IP and destination port, together with the protocol context. Two clients can connect to the same server port because their connection tuples differ. UDP demultiplexing depends on socket binding and implementation; a basic unconnected UDP server can receive datagrams from many senders on one local port.

Example: client A uses 192.0.2.10:51000 and client B uses 192.0.2.11:51000 to contact 198.51.100.20:443. The same source-port number does not make the connections identical because the source IP addresses differ. A single client can also open several connections using different local ports.

Remember: this transport use of multiplexing is distinct from dividing radio spectrum or time slots. Always explain which layer and resource the question concerns.

LO3 UDP operation and header

UDP sends independent datagrams without first establishing a transport connection. Its fixed header is eight bytes long and contains source port, destination port, length and checksum. The length includes header and payload. The checksum helps detect corruption using information from the UDP data and an IP pseudo-header; it does not correct damaged data or request a retransmission.

UDP offers no built-in ordering, duplicate suppression, retransmission or sender-rate regulation. Applications may supply these when needed. Its small header and lack of a transport handshake

can be useful for short exchanges and timing-sensitive applications. A datagram can still be delayed, dropped or reordered by the network.

Worked example: a UDP payload is 100 bytes. UDP length is 108 bytes. With a basic 20-byte IPv4 header, the IP packet is 128 bytes before link-layer overhead. This assumes no IPv4 options. A 40-byte IPv6 base header would instead give 148 bytes before any extension headers and link overhead.

For IPv4, a zero UDP checksum indicates omission; for IPv6, the UDP checksum is generally required, with limited protocol-specific exceptions. The core revision point is error detection, not reliability. DNS and real-time media are familiar UDP use cases, although their complete behaviour depends on their application protocols.

LO4 Reliable transfer and pipelining

Reliable transfer needs ways to detect corruption, identify data, acknowledge receipt and resend data after loss. Sequence numbers distinguish new data from retransmissions. Acknowledgements report progress. A timer lets the sender recover when a packet or its acknowledgement disappears. Receivers must handle duplicates so repeated transmissions do not become repeated application data.

Stop-and-wait sends one packet and waits for acknowledgement before sending another. It is simple but wastes capacity when round-trip time is large. Pipelining permits several outstanding packets. Go-Back-N uses a sender window and cumulative acknowledgements; in the basic model, receivers discard out-of-order packets and a timeout causes retransmission of the missing packet and later outstanding packets. Selective Repeat acknowledges and buffers packets individually and retransmits only missing ones, requiring more state and careful sequence-number handling.

Example: packets 1, 2, 3 and 4 are sent, but 2 is lost. A basic Go-Back-N receiver cannot accept 3 and 4 as in-order data, so the sender later repeats 2, 3 and 4. Selective Repeat can retain 3 and 4 and request or trigger recovery of 2 only.

Worked utilisation example: a 1,000-byte packet on 10 Mb/s takes 0.8 ms to transmit. With a 100 ms RTT and negligible acknowledgement transmission time, stop-and-wait utilisation is approximately $0.8/(100 + 0.8) = 0.00794$, or 0.79%. Pipelining keeps more data in flight and makes much better use of a long-delay link.

LO5 TCP connection and reliability

TCP establishes a connection with a three-way handshake. The client sends SYN with an initial sequence number, the server sends SYN plus ACK, and the client acknowledges. This confirms bidirectional reachability and synchronises initial sequence information. TCP headers include ports, sequence and acknowledgement numbers, flags, a receive-window field and checksum information.

TCP sequence numbers count bytes, not simply packets. An acknowledgement normally states the next byte expected. If data starts at byte 1,000 and contains 500 bytes, the next expected byte is 1,500, so an acknowledgement of 1,500 confirms receipt through byte 1,499. SYN and FIN each consume sequence-number space. A lost segment can be recovered by timeout or loss-detection mechanisms, including duplicate acknowledgements in classic TCP.

Flow control protects the receiver. The receiver advertises available buffer capacity as rwnd, and the sender limits outstanding data accordingly. Congestion control protects the network using a separate congestion window, cwnd. Sending is constrained by both windows and other implementation factors.

Example: a fast server sends to a phone with limited receive buffers. A small rwnd slows the server even if the network is uncongested. If the path is congested, cwnd may limit sending even when receiver buffers are large. TCP checksums detect corruption but do not encrypt data or authenticate the sender.

LO6 Congestion and TCP control mechanisms

Congestion occurs when traffic demand exceeds available network resources, such as link capacity or buffering. Symptoms include growing queues, high delay and packet drops. Retransmissions consume additional capacity, and a packet dropped late in its path has already used upstream resources. Increasing buffer size alone can create excessive delay, often called bufferbloat, without increasing the link's service rate.

In classic TCP, slow start grows cwnd approximately exponentially across RTTs as acknowledgements arrive. Despite its name, it can grow rapidly. On reaching the slow-start threshold, congestion avoidance grows more cautiously. With Reno-style behaviour, this is approximately additive growth. Loss triggers a reduction; a timeout generally produces a stronger response than recovery through duplicate acknowledgements. Fast retransmit can resend likely missing data without waiting for the timer, and fast recovery manages the window during recovery.

Example: a simplified slow-start window grows from 1 to 2 to 4 to 8 maximum-sized segments over successful rounds. These numbers illustrate the mechanism, not a universal initial-window setting. If all users increase aggressively on a constrained link, queues and loss signal the need to reduce rates.

Explicit Congestion Notification lets capable routers mark congestion instead of relying only on drops. Supporting endpoints report and react to these marks. Exact control behaviour depends on the TCP algorithm; modern implementations do not all follow Reno's window growth rules.

LO7 Fairness and transport evolution

Fairness concerns how competing flows share a constrained resource. Two similar long-lived TCP flows with comparable paths may tend toward a roughly even share under classic assumptions. Real results depend on RTT, congestion algorithm, number of connections, traffic pattern and policy. A user opening ten flows can receive more aggregate capacity than a user opening one, so fairness per flow is not necessarily fairness per person.

Network-assisted congestion control uses feedback from network devices, such as ECN marking, to inform endpoints. End-to-end approaches infer conditions from acknowledgements, loss and timing. Administrators may also use scheduling and traffic policies to protect important services; that policy choice differs from transport's default sharing behaviour.

Example: two equal flows on a 20 Mb/s bottleneck might each average near 10 Mb/s in an idealised case. A short-RTT flow or different algorithm can change that result. UDP applications that ignore congestion can crowd out responsive TCP traffic, so responsible UDP-based protocols need congestion handling.

Evolution includes TCP variants such as Reno and CUBIC, model-based approaches such as BBR, and QUIC over UDP. QUIC combines transport and security functions and supports multiple streams. Treat future directions as design tradeoffs involving latency, fairness, security and adaptability, rather than assuming one protocol replaces all others.

Unit 3 practice

1. A TCP segment starts at sequence number 2,000 and carries 300 bytes. What acknowledgement follows if all earlier bytes have arrived?

2. Explain flow control versus congestion control using a slow receiver and a busy router.

3. Packet 3 is lost in a four-packet window. Compare basic Go-Back-N and Selective Repeat recovery.

Unit 3 answer guidance

1. ACK 2,300: the next expected byte. This assumes a data segment and no extra SYN or FIN sequence consumption.

2. Flow control responds to receiver buffer capacity through rwnd. Congestion control responds to path conditions through cwnd. Either can restrict the sender independently.

3. Go-Back-N retransmits 3 and outstanding later packets, including 4. Selective Repeat can buffer 4 and retransmit 3 alone. State the basic receiver assumptions.

4 The network layer data plane

LO1 Forwarding routing and router components

Forwarding is the local action of moving a packet from an incoming interface to the selected outgoing interface. Routing determines the paths and information used to construct forwarding tables. A router's input ports perform reception, link processing and lookup; a switching fabric transfers packets internally; output ports queue and transmit them. Router ports here mean interfaces, not TCP or UDP application ports.

Destination-based IP forwarding uses the destination address and longest-prefix match. If several entries match, select the most specific prefix. A default route, /0, is used when no more specific route matches. A Layer 2 Ethernet switch instead normally forwards using destination MAC addresses; router switching fabric is an internal mechanism, not the same thing as an Ethernet LAN switch.

Worked example: a table contains 10.0.0.0/8 via A, 10.2.0.0/16 via B and 10.2.3.0/24 via C. A packet for 10.2.3.9 matches all three and uses C because /24 is longest. A packet for 10.2.8.9 uses B.

Example fault: a packet reaches a router but has no matching specific route and no default route. The router cannot forward it toward its destination. Changing the application's port will not create the missing route.

LO2 Traditional control plane architecture

In a traditional distributed control plane, routers run routing protocols and exchange information. Each calculates or selects routes, then installs forwarding information locally. The control-plane software makes decisions; the data plane applies those decisions to packets. OSPF and BGP serve different routing contexts and are studied further in Unit 5.

Benefits include distributed operation, mature protocols and the ability to adapt to topology changes. Limitations include configuration complexity, convergence delay and difficulty applying a coordinated policy across many devices. Devices can temporarily have inconsistent information after a failure, causing black holes or loops until convergence.

Example: a campus link fails. Routers detect the failure, exchange updated routing information, calculate alternative paths and update forwarding entries. Packets may be lost during this process even though the network eventually finds another route. A backup link helps only if routing and failure detection can use it.

Exam approach: describe where decisions happen, how information spreads, and what happens during change. Traditional routing is not necessarily static routing; a distributed control plane can run dynamic protocols.

LO3 Traditional networking and SDN

Software-Defined Networking separates forwarding behaviour from control logic through programmable interfaces. An SDN controller has a logically central view of selected network state and policies, then installs rules in switches or routers. Logically central does not mean there must be only one physical controller: redundant or distributed controller implementations are possible.

Traditional networking often couples device-specific control logic and forwarding inside each router. SDN can make coordinated policy, automation and experimentation easier. It also creates requirements for controller resilience, secure control channels, correct applications and reliable state management. Distributed protocols and SDN may coexist in a hybrid network.

Example: an institution wants guest traffic blocked from finance servers across 30 switches. A controller application can derive and distribute consistent rules instead of relying entirely on separate manual edits. If an incorrect policy is distributed, the impact can also be widespread, so testing and rollback matter.

Compare using decision location, configuration method, visibility, flexibility and failure handling. Do not say SDN removes physical devices or all distributed behaviour. The data plane still needs to forward traffic efficiently, and controller reachability and rule lifetime affect failure behaviour.

LO4 Network service models

A network service model describes what the network promises to applications and transport protocols. Possible promises include guaranteed delivery, bandwidth, maximum delay, bounded delay variation and in-order delivery. The Internet's basic IP service is best effort: it attempts delivery but does not guarantee any of those properties by itself.

Different applications need different properties. A file transfer needs completeness more than a strict short deadline. A live class needs low delay and manageable jitter, the variation in packet delay. A critical control application may require bounded delay and high availability. Quality-of-service mechanisms can prioritise or reserve resources within an administered network, but end-to-end guarantees require support across the relevant path.

Example: a campus gives live teaching traffic priority over bulk updates. This can improve campus queue behaviour under contention, but the campus cannot promise the same treatment across every external ISP and a student's home Wi-Fi. TCP can recover loss, yet it does not make IP a guaranteed-delay service.

Evaluation: identify the required service, available mechanisms and limits of administrative control. Reliability, throughput and delay are related performance concerns but are not interchangeable guarantees.

LO5 Queues buffers and scheduling

Packets can queue at input ports, output ports or inside a router. Buffers absorb temporary bursts, but persistent arrival rates above service rates eventually cause drops or sustained delay. Input queuing can suffer head-of-line blocking: the packet at the front waits for a busy output while packets behind it could otherwise use free outputs.

Scheduling decides which queued packet is served next. FIFO uses arrival order. Priority scheduling serves higher-priority queues first but can starve lower priorities without safeguards. Round robin gives queues turns. Weighted fair scheduling allocates service according to configured weights under suitable assumptions. Buffer management decides what to retain or drop; scheduling and buffer management are different decisions.

Worked example: two continually backlogged queues share a 12 Mb/s output with ideal weights 2:1. Their intended shares are 8 Mb/s and 4 Mb/s. If one queue is idle, a work-conserving implementation may let the other use the spare capacity. Actual packet sizes and algorithm details affect short-term results.

Example: huge FIFO buffers on an uplink keep packets from dropping immediately but make calls lag behind downloads. Appropriate queue management and prioritisation may reduce delay. Increasing memory alone does not increase output-link capacity.

LO6 IPv4 IPv6 and subnetting

IPv4 addresses are 32 bits; IPv6 addresses are 128 bits. A prefix identifies the network portion. IPv4 headers include source and destination addresses, TTL, protocol, length and checksum fields. TTL is reduced at each routed hop. The IPv6 base header is 40 bytes, uses a hop limit and has no base-header checksum; extension headers carry optional information. IPv6 routers do not fragment packets in transit; IPv4 fragmentation can occur when allowed.

For ordinary IPv4 subnets, host bits = 32 minus prefix length, total addresses = 2 to the power of host bits, and usual usable hosts = total minus 2. The network and directed-broadcast addresses account for the two exclusions. /31 point-to-point and /32 host routes are special cases, so do not apply minus 2 blindly.

Worked example: split 192.168.10.0/24 into four equal subnets. Borrow two bits, giving /26 and mask 255.255.255.192. Each block has 64 addresses and normally 62 usable hosts. Blocks start at .0, .64, .128 and .192. In 192.168.10.64/26, network = .64, usable = .65 to .126, broadcast = .127. Host .70 belongs here because it lies within that block.

Address planning example: a LAN needs 50 hosts. Six host bits provide 64 addresses and 62 usable, so /26 is sufficient. For 70 hosts, use seven host bits and /25, giving 126 usual usable hosts. Allocate non-overlapping ranges, document gateways and reserve appropriate infrastructure addresses. DHCP can distribute configured settings; private IPv4 ranges often use NAT for Internet access.

IPv6 notation uses hexadecimal groups; 2001:db8:1::10 is a documentation example. Leading zeroes may be omitted and one consecutive run of zero groups may use ::. Many IPv6 LANs use /64; IPv6 has no broadcast. Transition approaches include dual stack, tunnelling and translation such as NAT64. A host does not convert all IPv4 and IPv6 traffic automatically without suitable support.

LO7 Generalised forwarding and middleboxes

Generalised forwarding matches more packet fields than destination IP alone. Rules can inspect selected header fields such as source and destination addresses, transport ports and VLAN information, then apply actions such as forward, drop, modify or send to a controller. This is often described as match plus action. Rule priority resolves overlapping matches.

OpenFlow is one protocol for communication between a controller and supported forwarding devices. The controller can install flow entries; switches match packets against tables, apply actions and keep counters. A table miss follows the configured behaviour, which can include dropping or sending information to the controller. Not all SDN systems use OpenFlow.

Example rule: traffic from the guest VLAN to the finance subnet is dropped; approved traffic to an Internet gateway is forwarded. A more specific rule can take priority over a broad permit rule. Counters help determine whether the rule is actually being used.

Middleboxes perform functions beyond ordinary forwarding, such as firewall inspection, NAT, load balancing or intrusion detection. They can improve security and resource use but also add processing delay and state-related failure modes. NAT changes addressing and sometimes ports;

it is not a substitute for a complete security policy.

Unit 4 practice

1. Which route wins for 172.16.4.7 when /16 and matching /24 routes exist?

2. Find network, broadcast and usable range for 192.168.20.130/26.

3. Recommend queue treatment for live classes sharing an uplink with backups.

Unit 4 answer guidance

1. The matching /24 wins through longest-prefix matching.

2. /26 blocks start at 0, 64, 128 and 192. Network = 192.168.20.128; broadcast = .191; usable = .129 to .190. The host .130 is valid in this subnet.

3. Give live traffic appropriate priority or weighted service, prevent starvation of other traffic and schedule or limit backups. Check the actual bottleneck before changing the policy.

5 The network layer control plane

LO1 Control plane functions

The control plane creates and maintains information that governs forwarding. It discovers topology or reachability, calculates paths, selects routes and distributes rules. The data plane then handles individual packets using installed state. Management sets intended configuration and monitors operation; although it interacts with control, it is a separate responsibility.

Traditional routers coordinate through distributed protocols. An SDN controller can calculate policies and install rules through programmable interfaces. Both approaches need current information, failure handling and consistent state. A functioning route calculation is not enough if the forwarding hardware has not received the resulting entry.

Example: after a link fails, the control plane selects an alternative next hop. The data plane subsequently sends arriving packets through that interface. An administrator viewing logs and changing a routing policy is performing a management task that affects those decisions.

Exam answer: distinguish deciding from doing. 'Routing calculates or selects paths; forwarding applies a local next-hop decision to each packet' is a useful starting point, followed by the scenario's specific failure or policy requirement.

LO2 OSPF and BGP

An autonomous system is a network or collection of networks under a common routing administration. Intra-domain protocols operate within such a domain; inter-domain routing exchanges reachability between domains. OSPF is a link-state intra-domain protocol. Routers exchange link-state information, build a topology view and calculate least-cost paths. OSPF areas support hierarchy and scalability, with a backbone connecting the area structure.

BGP is a path-vector protocol used to exchange routes between autonomous systems and distribute BGP information within them. External BGP exchanges routes between ASes; internal BGP distributes information within an AS. Route selection depends on attributes and policy, not simply physical distance. AS_PATH records traversed ASes and supports loop prevention; other attributes help express preferences.

Example: a university can use OSPF to choose paths between campus buildings, while its Internet connectivity may depend on BGP arrangements with providers. OSPF's 'shortest' means lowest configured cost, not the fewest kilometres. BGP can prefer a commercially suitable path even if another has a shorter AS path.

Remember: BGP is not a protocol that measures and always picks the fastest Internet path. Explain scope, exchanged information and selection criteria.

LO3 Link-state and distance-vector algorithms

Link-state routing floods information about local links so routers can construct a topology map and run a shortest-path algorithm such as Dijkstra. Distance-vector routing exchanges estimates of destination cost with neighbours and updates them using the Bellman-Ford principle: choose the minimum of cost to a neighbour plus that neighbour's advertised cost to the destination.

Worked example: A connects to B with cost 2 and to C with cost 7. B connects to C with cost 1. A can reach C directly for 7 or through B for 2 + 1 = 3, so the lower-cost path is A to B to C. Link-state reaches this result from a topology map; distance-vector can reach it from B's advertised distance to C.

Link-state requires topology storage and flooding overhead but can converge efficiently with complete information. Distance-vector is conceptually simpler and shares neighbour estimates, but some forms suffer loops and count-to-infinity after failures. Techniques such as split horizon and poisoned reverse help in certain situations but do not solve every possible loop.

OSPF illustrates link-state. RIP is a familiar distance-vector example. BGP is path-vector, so do not casually classify it as ordinary distance-vector without discussing its path information and policy mechanisms.

LO4 Routing policy and address management

Routing policy expresses which routes a network accepts, prefers and advertises. It reflects security, business relationships and traffic-engineering goals. A provider may agree to carry a customer's traffic while declining to provide free transit between unrelated peers. Import and export filters constrain these decisions.

Address management requires unique allocation, documented ownership and appropriate aggregation. CIDR prefixes let a network advertise a block rather than every individual subnet. Aggregation reduces routing-table entries but must reflect actual allocation and reachability. Public Internet presence involves valid address resources or provider connectivity, DNS, routing and access controls; configuring a private address on a server does not make it publicly reachable.

Example: an institution has 10.40.0.0/16 internally and allocates /24 subnets to departments. It can summarise internally where appropriate. This private prefix should not be advertised as a public Internet destination. Public services might use provider-assigned public addresses with controlled inbound access, or hosted infrastructure.

Security example: filter unexpected route advertisements and validate intended prefixes to reduce accidental route leaks. Evaluate a policy using who should reach whom, through which provider, and what traffic the organisation is willing to carry.

LO5 Unicast multicast and anycast

Unicast delivers to one addressed endpoint. Multicast delivers to members of a subscribed group using supporting network mechanisms. Anycast allows multiple service instances to advertise the same destination address; routing directs a packet toward one instance according to the network's routing decisions. Anycast normally delivers to one selected instance, not every instance.

Example: a distributed DNS service has instances in several regions advertising one anycast address. A user's network routes queries toward a reachable preferred instance. If an instance withdraws its route, traffic can shift to another. This improves distribution and resilience, but the chosen instance is not necessarily the geographically closest or least loaded.

Multicast could distribute one stream efficiently to many group members within a supporting network. Unicast instead creates separate delivery relationships. Many public video services use unicast HTTP delivery through CDNs rather than relying on Internet-wide multicast.

Tradeoff: a route change during a long stateful session may send packets to an instance without the original session state. Anycast service design must account for this. Compare recipient count, addressing and routing behaviour rather than treating these methods as interchangeable.

LO6 SNMP NETCONF and YANG

Network management involves measuring health, detecting faults, managing configuration and assessing capacity and security. A management station communicates with managed devices or their agents. Metrics can include interface state, packet counters, errors, utilisation and CPU load. Baselines help distinguish normal activity from unusual conditions.

SNMP uses managed objects organised through management information bases, or MIBs. A manager can query values; agents can send notifications such as traps or informs. Counter differences over time reveal rates. SNMPv3 supports stronger authentication and privacy features when appropriately configured. Older community-string approaches need careful protection.

NETCONF exchanges structured configuration operations, commonly using XML over a secure transport such as SSH. It supports capabilities and configuration datastores according to the device's implementation. YANG models define the structure, types and constraints of configuration and operational data. YANG is a modelling language, not the transport that sends configuration.

Example: monitor a switch's uplink octet counter using SNMP, then use a supported NETCONF operation to change a setting whose schema is defined in YANG. If the counter rises by 15,000,000 bytes in 10 seconds, average traffic is 15,000,000 x 8 / 10 = 12 Mb/s. Check direction, counter width and measurement interval. Protect management access and record changes.

LO7 ICMP diagnostics and error reporting

ICMP supports network-layer control messages, including selected errors and diagnostic exchanges. Ping commonly sends Echo Request and receives Echo Reply to test reachability and estimate round-trip time. An unreachable response can explain why delivery failed. Time Exceeded indicates that a packet's TTL or hop limit expired, among other defined conditions.

Traceroute sends probes with progressively increasing TTL values. Routers where TTL expires may return Time Exceeded messages, allowing the tool to infer hops. Implementations may use UDP, ICMP or TCP probes. A router can rate-limit or block replies, and forward and return paths can differ, so missing output does not automatically prove a failed forwarding device.

Example: the local gateway answers ping but the portal name does not resolve. This suggests investigating DNS rather than assuming the gateway is down. If the destination does not answer ping but HTTPS works, its policy may block Echo messages while allowing web traffic. Ping alone cannot verify the complete application service.

ICMP does not repair every network problem or guarantee delivery. Some messages are important for functions such as path MTU discovery, so blocking all ICMP can create problems. Use evidence from several tests and understand exactly what each test measures.

Unit 5 practice

1. In a graph, A-B costs 4, A-C costs 9 and B-C costs 2. Find A's lower-cost path to C.

2. Explain why an ISP may choose a BGP route that is not the shortest in AS hops.

3. A host does not respond to ping but its web page loads. What can you conclude?

Unit 5 answer guidance

1. Through B costs 4 + 2 = 6, lower than direct cost 9.

2. Route policy and attributes can express commercial or administrative preferences. AS path length is only one consideration in selection.

3. The web service is reachable by the tested method. ICMP Echo may be blocked or limited. Do not conclude that the host is offline solely from failed ping.

6 The link layer and local area networks

LO1 Link layer services and implementation

The link layer transfers frames between adjacent network nodes across one link. It provides framing, medium access and error detection; some link protocols also offer reliable local delivery or flow control. These services differ by technology. Ethernet usually detects and discards damaged frames without providing Ethernet-level retransmission; Wi-Fi can acknowledge and retry local frames.

Implementation involves network interfaces, device drivers and protocol processing. A LAN can use Ethernet or Wi-Fi, while point-to-point WAN links use suitable link technologies. A broadcast medium is shared among several potential receivers; a point-to-point link connects two endpoints. Physical transmission still carries the bits below the link-layer functions.

Example: a laptop sends an IP packet toward a router over Wi-Fi. The Wi-Fi link carries it in a frame. The router later sends the IP packet over another link in that link's frame format. The packet's local delivery service changes with the link technology.

Remember: 'link layer' is not a universal guarantee of reliable end-to-end delivery. Local retry on one wireless hop does not replace transport recovery across the complete route.

LO2 How the link layer supports the other layers

At the sender, the link layer encapsulates a network-layer packet in a frame. At the next node, link processing checks the frame and passes the packet upward as appropriate. At a router, the incoming frame ends there; a new frame is created for the outgoing link. At the destination host, data progresses through IP, transport and application processing.

Example: a campus laptop with address 192.168.1.10 sends to a remote server. The Ethernet frame's destination MAC is the local gateway's MAC, while the IP destination remains the remote server's address. The router forwards the packet using a new frame addressed to the next local hop. Without NAT or tunnelling, endpoint IP addresses remain the same while TTL and related header information change.

Layer relationship: physical carries signals, link organises local frames, network connects different networks, transport reaches processes and application interprets the messages. In OSI, link is Layer 2 and network is Layer 3. MAC addresses do not replace IP routing across the Internet.

Troubleshooting example: if a device cannot reach its local gateway, examine local link state, VLAN placement, addressing and neighbour resolution before blaming a remote application server.

LO3 Parity checks checksums and CRC

Error detection adds redundant information so a receiver can detect certain changes in transmitted bits. Detection is different from correction: noticing damage does not automatically reconstruct the original data. No finite detection code detects every conceivable error pattern.

Parity adds a bit to make the number of 1 bits even or odd. With even parity, data 1011001 has four 1 bits, so append 0. A single flipped bit changes parity and is detected. Two flipped bits may leave parity unchanged. Two-dimensional parity can detect more patterns and locate some

single-bit errors.

A checksum combines data values into a check value; Internet-style checksums use one's-complement arithmetic. CRC treats bit strings as polynomials, divides by an agreed generator using modulo-2 operations and transmits a remainder. The receiver checks consistency with the generator. Properly chosen CRCs are especially effective against specified burst-error patterns; capabilities depend on the generator and frame length.

Worked CRC example: data 1101 with generator 1011 has a three-bit remainder because the generator degree is three. Append three zeroes: 1101000. Modulo-2 division gives remainder 001. Transmit 1101001. Dividing this codeword by 1011 gives zero remainder. Use XOR rather than ordinary subtraction with borrowing. A zero remainder indicates no detected error, not mathematical proof that no error occurred.

LO4 Sharing a medium

Multiple access protocols coordinate transmitters sharing one communication medium. Channel partitioning assigns time, frequency or code resources. It avoids certain contention problems but can leave allocations unused. Random access lets devices attempt transmission as needed, with mechanisms to handle competing attempts. Taking-turns approaches coordinate use through polling or a token.

ALOHA illustrates random access with retransmission after collisions. CSMA listens before transmitting, but propagation delay can still allow simultaneous attempts. Classic shared half-duplex Ethernet uses CSMA/CD, detecting collisions while transmitting. Modern switched full-duplex Ethernet links do not use CSMA/CD for normal operation. Wi-Fi uses collision avoidance mechanisms, including contention and backoff, because detecting a collision while transmitting radio is difficult.

Example: two wireless learners both find the channel idle and try to send. Backoff and acknowledgement procedures help manage contention and retry unsuccessful transmissions. Hidden terminals may fail to hear each other even though both interfere at the access point; RTS/CTS can help in some circumstances but adds overhead.

Compare efficiency at light and heavy load, coordination overhead and delay predictability. Do not apply an old shared-Ethernet collision model to every modern Ethernet LAN.

LO5 ARP Ethernet and switching

Ethernet frames carry destination and source MAC addresses, a type-related field, payload and a frame check sequence. A switch learns source MAC addresses and their incoming ports. For a known destination, it forwards on the learned port. For an unknown unicast destination, it normally floods eligible ports within the VLAN except the incoming port. Learned entries age out over time.

ARP resolves a local IPv4 next-hop address to a MAC address. If the destination is on the same subnet, the host resolves that destination. If it is remote, the host resolves its gateway, not the remote server's MAC across the Internet. An ARP request is normally broadcast locally; a reply is typically unicast. IPv6 uses neighbour discovery rather than ARP.

Example: 192.168.1.10/24 wants to send to 192.168.1.20/24. It asks who has .20, obtains that host's MAC and sends a frame there. To reach 203.0.113.20, it instead finds the gateway's MAC and sends the remote-destination IP packet in a locally addressed frame.

Security application: ARP messages can be spoofed because basic ARP lacks authentication. Segmentation and suitable switch protections can reduce exposure. Always distinguish the switch's MAC table, the host's ARP cache and the router's IP forwarding table.

LO6 VLANs and data centre architecture

A VLAN creates a logical Layer 2 broadcast domain on switching infrastructure. Devices in separate VLANs are separated at Layer 2 even if connected to the same physical switch. An access port normally carries one assigned VLAN for an endpoint. A trunk carries multiple VLANs, commonly using IEEE 802.1Q tags. Inter-VLAN communication needs Layer 3 routing and should be governed by access policy.

Example: put students in VLAN 10 and administration in VLAN 20. Student broadcasts remain in VLAN 10. A router or Layer 3 switch can permit selected student access to a shared learning server while blocking access to payroll. Creating VLANs alone does not prevent all communication if inter-VLAN routing is broadly permitted.

Data centres host applications across servers, storage and network devices. Virtualisation allows multiple workloads on shared hardware; virtual switches extend networking to virtual machines. Redundant links and equipment improve availability. Leaf-spine designs connect leaf switches through spine switches to provide multiple paths and support east-west traffic between servers. Load balancers distribute application requests.

Evaluation: segmentation limits broadcast scope and supports policy, while capacity and redundancy address different needs. More VLANs do not automatically create more bandwidth. Plan routing, trunk membership, address ranges and failure handling together.

LO7 DHCP UDP and DNS in host communication

A newly connected host needs an address, subnet information, an appropriate gateway and usually a DNS resolver. DHCPv4 can supply these through Discover, Offer, Request and Acknowledge exchanges, often remembered as DORA. A lease makes allocation time-limited. DHCP servers use UDP port 67 and clients use port 68. A relay can carry DHCP messages between a client subnet and a server on another network.

After configuration, the host identifies whether a destination is local or remote, resolves the local next-hop MAC through ARP and uses DNS when communicating by name. A DNS query commonly uses UDP port 53, though DNS also uses TCP and other secured transports. UDP carries application messages but does not itself assign addresses or resolve names.

Example: a student joins campus Wi-Fi, receives 192.168.10.25/24, gateway .1 and a resolver from DHCP, resolves the portal's name, resolves the gateway MAC for a remote portal and sends application traffic. If no lease is received, inspect DHCP reachability, VLAN and relay configuration. If the gateway works but names fail, inspect DNS settings.

Layer reminder: DHCP and DNS are application protocols, UDP is transport, IP is network and ARP/Ethernet support local delivery. They appear together here because a working LAN depends on their cooperation. DHCPv6 and IPv6 SLAAC differ from the DHCPv4 DORA process.

Unit 6 practice

1. A host sends to a server on another subnet. Whose MAC address belongs in the first frame's destination field?

2. Explain how VLANs separate student and payroll devices, and what additional control is needed.

3. A host has a valid lease and can reach its gateway, but website names fail. What should you test next?

Unit 6 answer guidance

1. The local next-hop router or gateway MAC. The IP packet still identifies the remote server as its destination, subject to later translation if used.

2. VLANs create separate broadcast domains. Inter-VLAN routing needs access controls to allow required services and block unauthorised access.

3. Verify the configured DNS resolver, test name resolution and check reachability of the resolver. A valid lease does not guarantee correct DNS settings.

Putting the units together

A complete student portal example

A learner joins campus Wi-Fi and opens a remote learning portal. First, the wireless link must connect successfully and any required access authentication must complete. DHCPv4 can provide the IPv4 address, mask, gateway and DNS resolver. The host uses the mask to recognise that the portal is outside its local subnet.

The browser asks DNS for the portal's address; cached results may avoid a fresh network exchange. For a typical HTTPS connection using TCP, the host establishes TCP and TLS before exchanging protected HTTP messages. Other web transport arrangements exist, so identify the protocol assumed by a question.

The host places transport data inside an IP packet. For the remote destination, it uses ARP to obtain the local gateway MAC if necessary. The wireless frame carries the packet to the local infrastructure. Switches forward local frames, while routers perform IP forwarding according to tables produced by control-plane decisions.

At each routed hop, the incoming frame is removed and an outgoing link frame is created. Routers may queue packets. If packets are lost, TCP can retransmit; if congestion is detected, its sending rate can fall. A CDN or cache may serve suitable portal content close to the user. The destination decapsulates received data and the application responds.

Security is applied across this process: authenticated access, protected sessions, restricted VLAN routing, hardened devices and monitored management. One measure cannot replace all the others.

A systematic troubleshooting method

1. Define the symptom precisely: one device or many, one application or all traffic, constant or intermittent. Record what changed and when.

2. Check physical and link conditions: cables, power, signal, port state, access-point connection and VLAN assignment.

3. Verify IP settings: address, mask or prefix, duplicate-address symptoms, gateway and DHCP lease. An address in the wrong subnet can prevent local communication.

4. Test local reachability and neighbour resolution. Check the gateway and relevant ARP or neighbour entries. A successful ping tests only the permitted ICMP exchange.

5. Check routes and remote connectivity. Use routing information and diagnostic probes, while accounting for filtering and asymmetric paths.

6. Test DNS separately from IP connectivity. Then check application ports, server responses and authentication.

7. Measure performance: latency, loss, utilisation, throughput and queue behaviour. Identify the bottleneck before selecting a remedy.

8. Make a controlled change, verify the result and document it. Retest the original failing activity, not only the diagnostic command.

Example: the portal works for staff but not students. Check student VLAN routing and access policy before assuming a global portal outage. If both groups load the portal but video stalls only on Wi-Fi, inspect wireless conditions and airtime.

Applying network security across the module

Use least privilege: permit required traffic and restrict unnecessary services. Separate guest, student, staff, server and management networks where justified. Harden routers and switches with secure administration, updates, protected credentials and backups. Protect controller and management access because changes there affect many devices.

Use encrypted services where appropriate, including HTTPS and secure management. TCP reliability and CRC error detection do not provide confidentiality. Configure logging and monitoring to identify unexpected traffic or configuration changes. Document baseline behaviour so anomalies can be investigated.

In practical work, use the authorised lab environment. Scan only lab targets you are permitted to test. Explain evidence in context: a port response identifies an observed service condition, not automatically a vulnerability or proof of safety.

Worked scenario: students need the portal but must not access payroll. Put the groups in suitable VLANs, route through controlled Layer 3 interfaces and allow only approved service traffic. Verify both an allowed connection and a blocked connection. Continue to authenticate portal users because network location alone is insufficient access control.

Calculation reference and extra worked examples

Bits = bytes x 8. Use decimal networking rates unless the question specifies another convention: 1 Mb/s = 1,000,000 bits/s. When a size is stated in MiB, use 1 MiB = 1,048,576 bytes. State your assumption if the question is ambiguous.

Transmission delay = packet bits / link bits per second. Propagation delay = distance / propagation speed. Convert seconds to milliseconds by multiplying by 1,000. Ideal transfer time = total bits / useful rate. Path bottleneck rate is the minimum link rate under a simple single-flow model.

Worked multi-hop delay: three store-and-forward links each operate at 2 Mb/s. A 1,000-byte packet takes 8,000/2,000,000 = 4 ms per link. With 2 ms propagation per link and no processing or queuing delay, total is 3 x (4 + 2) = 18 ms for the complete packet. This concerns one packet; a stream can be pipelined.

Worked file transfer: a 25 MB file at 10 Mb/s needs at least 25 x 1,000,000 x 8 / 10,000,000 = 20 seconds under decimal units. At an achieved useful rate of 8 Mb/s it needs 25 seconds. Use goodput when the question specifies useful file delivery.

Worked subnet requirement: 120 ordinary IPv4 hosts need at least 122 total addresses to allow network and broadcast. Seven host bits give 128 total and 126 usable, so choose /25. A /26 gives only 62 usable and is too small.

Worked subnet boundaries: /27 means 32-address blocks and mask 255.255.255.224. In 192.168.5.96/27, network is .96, usable range .97 to .126 and broadcast .127. Eight /27 blocks fit inside one /24.

Worked route: 192.168.0.0/16 goes to A, 192.168.5.0/24 to B and 192.168.5.96/27 to C. Destination 192.168.5.110 uses C. Destination .150 uses B. An address outside 192.168.0.0/16 uses another matching route, possibly the default.

Comparisons students must be able to explain

TCP versus UDP: reliable ordered byte stream with a connection versus independent datagrams without built-in delivery guarantees. Compare application needs and recovery cost.

Flow control versus congestion control: protect receiver buffers versus manage load on the network. Identify rwnd versus cwnd.

Routing versus forwarding: compute or select paths versus apply a next-hop rule to a packet.

MAC versus IP versus port: local link identifier versus network-layer address versus process endpoint identifier. They operate at different scopes.

Switch versus router: normally forward local Ethernet frames by MAC versus forward IP packets between networks. Layer 3 switches can also route.

DNS versus DHCP versus ARP: resolve names and other records versus distribute host configuration versus resolve a local IPv4 next-hop MAC.

Transmission versus propagation: putting packet bits on the link versus moving the signal across the link. Rate affects the first; distance and medium speed affect the second.

Error detection versus reliability versus security: detect certain corruption versus recover or manage delivery versus protect confidentiality, integrity and access. These are related but distinct functions.

Unicast versus multicast versus anycast: one endpoint versus group members versus one routed choice among service instances sharing an address.

Data plane versus control plane versus management: process packets versus determine forwarding state versus configure and observe the system.

Writing stronger examination answers

Define means provide the precise meaning. Describe means give characteristics or steps. Explain means show how or why something works. Compare means discuss both concepts using matching criteria. Analyse means break a problem into causes and relationships. Evaluate means judge options using criteria and reach a supported conclusion. Calculate means show formula, substituted values, units and result.

Weak answer: 'Use TCP because it is reliable.' Stronger answer: 'The assignment upload must arrive complete and in order. TCP uses sequence numbers, acknowledgements and retransmissions to recover missing data, which suits the file-transfer requirement. Recovery may delay completion, but missing content would be unacceptable.'

Weak answer: 'Use VLANs for security.' Stronger answer: 'Place student and payroll devices in separate VLANs to separate broadcast domains. Route between them through an access-controlled interface. Permit required portal access and deny unauthorised payroll access; test both outcomes.'

For a scenario, identify the requirement, explain the relevant mechanism, apply it to the stated facts and justify the result. Use the mark allocation to judge depth. Do not invent a universal rule that every mark equals one sentence. For calculations, marks may reward method as well as the final value.

Integrated practice scenario

A college has 80 student PCs, 20 staff PCs and a shared server. A 20 Mb/s Internet uplink carries live classes and large backups. Students sometimes fail to open the portal by name, although they can reach their gateway. Staff and students share one flat LAN. A new teaching building will be connected with a second routed path for resilience.

1. Recommend ordinary IPv4 subnet sizes for the 80 student PCs and 20 staff PCs, allowing a gateway in each. Give two non-overlapping example subnets.

2. Explain the first checks for the portal-name failure.

3. Explain how VLANs and routing policy can separate student and staff traffic while allowing portal access.

4. Analyse why adding uplink buffer memory may not solve live-class delay. Recommend two suitable actions.

5. Explain the different roles of forwarding, OSPF and monitoring when the primary building link fails.

6. Recommend transport properties for assignment upload and live audio, with reasons.

Integrated practice answer guidance

1. Students need at least 81 usable addresses including a gateway, so /25 provides 126. Staff need at least 21, so /27 provides 30, assuming no additional devices beyond stated needs. Examples: 192.168.50.0/25 for students, usable .1 to .126, broadcast .127; 192.168.50.128/27 for staff, usable .129 to .158, broadcast .159. Document reserved addresses and future growth.

2. Check the DHCP-provided resolver and test name resolution. Determine whether the issue affects one client or a whole VLAN. Verify DNS service reachability and responses. Gateway reachability demonstrates only part of the path and does not establish that the DNS server is reachable.

3. Put groups in separate VLANs and IP subnets. Provide controlled inter-VLAN routing. Permit authorised traffic to the portal and restrict sensitive staff services. Configure access and trunk ports correctly. Verify permitted and denied flows.

4. More memory can increase queueing delay without increasing the 20 Mb/s service rate. Use suitable queue management and class prioritisation, and schedule or limit backups. If measured demand remains above capacity, evaluate a capacity upgrade. Check whether Wi-Fi or the external path contributes to the problem.

5. OSPF exchanges changed topology information and calculates an alternative lower-cost available route. Forwarding applies the installed next-hop state to packets. Monitoring detects interface or traffic changes and records the outage and recovery. A redundant path helps only when it is usable and correctly configured.

6. Upload requires reliable complete delivery, favouring TCP or another suitable reliable transport. Live audio emphasises timely delivery, often using UDP-based media protocols with loss handling and congestion control. Security may require encryption in either case. Explain requirements rather than claiming one transport is universally superior.

Final readiness checklist

Can you explain every numbered outcome without simply reciting its title? Can you trace a request through all five teaching layers? Can you calculate delays and convert bytes to bits? Can you find an IPv4 subnet's network, usable range and broadcast? Can you explain a TCP acknowledgement and distinguish its two windows? Can you choose the longest matching route? Can you compare OSPF and BGP? Can you explain ARP, switching and VLAN routing? Can you propose a diagnosis supported by evidence? Can you justify a configuration and its security implications?

If an answer is no, revisit the relevant example and solve a modified version. Work through the prescribed reading and Arc activities for depth beyond this guide.

Source and reading note

Outcome coverage is based on The Independent Institute of Education, 2026, Computer Networks 1B CONE5112/w Module Outline, supplied by the lecturer, especially page 7 and pages 14 to 25. Outcome headings here paraphrase the outline; the explanations, calculations, scenarios and practice questions are newly written teaching material.

The outline prescribes Kurose, J. F. and Ross, K. W., 2022, Computer Networking A Top Down Approach, eighth Global Edition, Pearson. Consult that text and Arc for the full prescribed treatment. This guide does not claim to quote or reproduce textbook passages. Examples use private or documentation addresses and reserved example names.